Huawei Access Node Optical Transport Failure SOP (IND-ACC-58)

Node: IND-ACC-58 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_2329

Description

The Huawei Access Node Optical Transport Failure SOP (IND-ACC-58) is designed to address a specific failure domain on the Huawei Access Node at Indiranagar Exchange. This failure domain is characterized by the occurrence of Alarm Code ALARM_2329, indicating a warning-level Optical Transport Failure. This failure can be attributed to the degradation of SFP/XFP, fiber cuts, or Rx/Tx power loss. In the context of the Huawei Access Node, the Optical Transport Failure is primarily related to the 10G Ethernet over fiber link, which is utilized for the access network. The SFP/XFP module plays a crucial role in this link, as it enables the transmission and reception of 10G data over the fiber optic cable. The degradation of the SFP/XFP module or a fiber cut can lead to a loss of signal strength, resulting in the Optical Transport Failure. The Huawei Access Node, being a sophisticated networking device, employs a range of protocols and mechanisms to detect and respond to failures in the optical transport chain. The Device Management Entity (DME) and the Network Element Management System (NMS) work in conjunction to monitor the health of the optical transport links and provide real-time alarm notifications when a failure is detected. The Huawei Access Node's optical transport failure can be further attributed to the use of Wavelength Division Multiplexing (WDM) technology, which enables the multiplexing of multiple wavelengths on a single fiber optic cable. In the event of a failure, the WDM system can experience signal degradation or loss, leading to the Optical Transport Failure. This SOP is designed to provide a step-by-step guide for resolving Optical Transport Failures on the Huawei Access Node at Indiranagar Exchange, with a focus on the root causes, diagnostic commands, resolution steps, verification commands, and rollback procedures.

Pre-requisite Checks

- Ensure that the remote hands are dispatched to Indiranagar Exchange to gain physical access to the Huawei Access Node.
- Verify that the redundant link status is checked to ensure that it is functioning correctly.
- Confirm that the Huawei Access Node is powered on and that the system is stable.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display fault alarm all, Expected Output: Alarm Code: ALARM_2329, Alarm Level: Warning, Alarm Message: Optical Transport Failure, Description: This command isolates the root cause of the Optical Transport Failure by displaying all active alarms on the Huawei Access Node, including the Alarm Code, Alarm Level, and Alarm Message.

Command: display interface gigabitethernet 0/0/0, Expected Output: Interface Status: Down, Reason: Optical Transport Failure, Description: This command isolates the specific interface affected by the Optical Transport Failure by displaying its status, including the interface number, status, and reason for the failure.

Command: display sfp xfp all, Expected Output: SFP/XFP Status: Down, Reason: Degradation, Description: This command isolates the SFP/XFP module affected by the Optical Transport Failure by displaying its status, including the SFP/XFP number, status, and reason for the failure.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace sfp xfp 0/0/0, Description: This step resolves the Optical Transport Failure by replacing the degraded SFP/XFP module with a new one. The Huawei Access Node will automatically detect the new module and reconfigure the optical transport link.

Step 2: Step: Repair the fiber cut, Command: repair fiber cut 0/0/0, Description: This step resolves the Optical Transport Failure by repairing the fiber cut. The Huawei Access Node will automatically detect the repaired fiber and reconfigure the optical transport link.

Step 3: Step: Adjust the Rx/Tx power, Command: adjust rxtx power 0/0/0, Description: This step resolves the Optical Transport Failure by adjusting the Rx/Tx power of the affected interface. The Huawei Access Node will automatically detect the adjusted power and reconfigure the optical transport link.

Post-Check Verification

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Command: display fault alarm all, Expected Output: Alarm Code: None, Alarm Level: None, Alarm Message: None, Description: This command verifies that the Optical Transport Failure has been resolved by displaying all active alarms on the Huawei Access Node, including the Alarm Code, Alarm Level, and Alarm Message.

Command: display interface gigabitethernet 0/0/0, Expected Output: Interface Status: Up, Reason: None, Description: This command verifies that the specific interface affected by the Optical Transport Failure has been resolved by displaying its status, including the interface number, status, and reason for the failure.

Command: display sfp xfp all, Expected Output: SFP/XFP Status: Up, Reason: None, Description: This command verifies that the SFP/XFP module affected by the Optical Transport Failure has been resolved by displaying its status, including the SFP/XFP number, status, and reason for the failure.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: restore sfp xfp 0/0/0, Description: This step safely rolls back the changes made to resolve the Optical Transport Failure by restoring the original SFP/XFP module. The Huawei Access Node will automatically detect the restored module and reconfigure the optical transport link.

- Step: Restore the original fiber configuration, Command: restore fiber configuration 0/0/0, Description: This step safely rolls back the changes made to resolve the Optical Transport Failure by restoring the original fiber configuration. The Huawei Access Node will automatically detect the restored fiber and reconfigure the optical transport link.

- Step: Restore the original Rx/Tx power, Command: restore rxtx power 0/0/0, Description: This step safely rolls back the changes made to resolve the Optical Transport Failure by restoring the original Rx/Tx power of the affected interface. The Huawei Access Node will automatically detect the restored power and reconfigure the optical transport link.